



**KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY
(AN AUTONOMOUS INSTITUTION)**

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SDC1-WEEK1-DAY1

II-I CSE, CSE(AI&ML)

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SLNo	Week		Detailed Subtopics		Detailed Subtopics
1	Week 1	Part A Introduction to Machine Learning	Definition and scope of ML, AI vs ML vs DL, Types of learning (Supervised, Unsupervised, Semi-supervised, Reinforcement), Real-world ML applications in healthcare, finance, vision	Part B Supervised Learning: Polynomial Regression	Introduction, Need, Equation, Features, Workflow, Degree Selection, Model Fitting, Underfitting & Overfitting, Advantages, Disadvantages, Applications

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(Week 1-Session 1 Part A: Introduction to Machine Learning)

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1. Introduction to Machine Learning

1.1 What is Machine Learning?

Machine Learning (ML) is a subfield of Artificial Intelligence that enables systems to learn patterns from data and make decisions without being explicitly programmed. Instead of hard-coding rules, ML algorithms infer relationships from examples.

ML is best described as data-driven decision making.

Classic definition (Tom Mitchell):

A computer program is said to learn from experience (E) with respect to some task (T) and performance measure (P), if its performance at task (T), as measured by (P), improves with experience (E).

- Task (T): Problem being solved (classification, regression)
- Experience (E): Data
- Performance (P): Accuracy, error, reward

1.2 Why Machine Learning is Needed

Traditional programming requires exact instructions and doesn't handle complex tasks like understanding images or language well. It can't efficiently process large amounts of data. Machine Learning solves these problems by learning from examples and making predictions without fixed rules. Let's see various reasons why it is important:

1. Solving Complex Business Problems

Traditional programming struggles with tasks like language understanding and medical diagnosis. ML learns from data and predicts outcomes easily.

Examples:

- Image and speech recognition in healthcare.
- Language translation and sentiment analysis.

2. Handling Large Volumes of Data

The internet generates huge amounts of data every day. Machine Learning processes and analyzes this data quickly by providing valuable insights and real-time predictions.

Examples:

- Fraud detection in financial transactions.
- Personalized feed recommendations on Facebook and Instagram from billions of interactions.

3. Automate Repetitive Tasks

ML automates time-consuming, repetitive tasks with high accuracy hence reducing manual work and errors.

Examples:

- Gmail filtering spam emails automatically.
- Chatbots handling order tracking and password resets.
- Automating large-scale invoice analysis for key insights.

4. Personalized User Experience

ML enhances user experience by tailoring recommendations to individual preferences. It analyzes user behavior to deliver highly relevant content.

Examples:

- Netflix suggesting movies and TV shows based on our viewing history.
- E-commerce sites recommending products we're likely to buy.

5. Self Improvement in Performance

ML models evolve and improve with more data helps in making them smarter over time. They adapt to user behavior and increase their performance.

Examples:

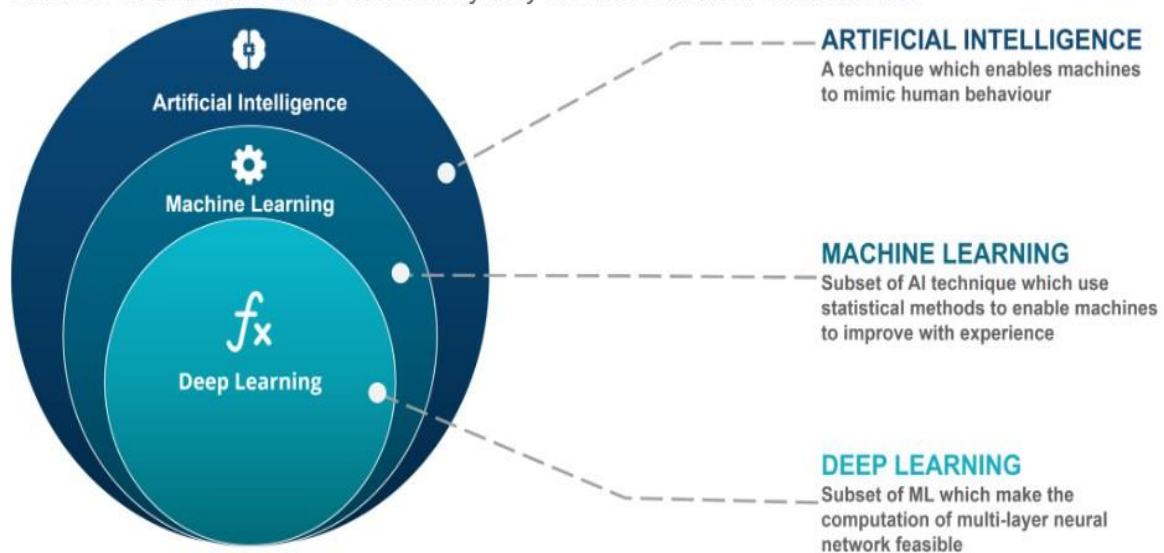
- Voice assistants like Siri and Alexa learning our preferences and accents.
- Search engines refining results based on user interaction.
- Self-driving cars improving decisions using millions of miles of driving data.

1.3 ML vs AI vs Deep Learning

Artificial Intelligence → Machine Learning → Deep Learning

“

Artificial Intelligence is the broader umbrella under which Machine Learning and Deep Learning come. And you can also see in the diagram that even deep learning is a subset of Machine Learning. So all three of them AI, machine learning and deep learning are just the subsets of each other. So let us move on and understand how exactly they are different from each other. ”



- **AI:** Broad goal of making machines intelligent
- **ML:** Data-driven learning
- **DL:** Neural-network-based ML with representation learning

1.4 What Makes a Machine "Learn"?

A machine "learns" by identifying patterns in data and improving its ability to perform specific tasks without being explicitly programmed for every scenario. This learning process helps machines to make accurate predictions or decisions based on the information they receive. Unlike traditional programming where instructions are fixed, ML allows models to adapt and improve through experience.

Here is how the learning process works:

1. **Data Input:** Machine needs data like text, images or numbers to analyze. Good quality and enough quantity of data are important for effective learning.
2. **Algorithms:** Algorithms are mathematical methods that help the machine find patterns in data. Different algorithms help different tasks such as classification or regression.
3. **Model Training:** During training, the machine adjusts its internal settings to better predict outcomes. It learns by reducing the difference between its predictions and actual results.
4. **Feedback Loop:** Machine compares its predictions with true outcomes and uses this feedback to correct errors. Techniques like gradient descent help it update and improve.
5. **Experience and Iteration:** Machine repeats training many times with data helps in

refining its predictions with each pass, more data and iterations improve accuracy.

6. **Evaluation and Generalization:** Model is tested on unseen data to ensure it performs well on real-world tasks.

Machines "learn" by continuously increasing their understanding through data-driven iterations like how humans learn from experience.

1.5 Importance of Data in Machine Learning

Data is the foundation of machine learning (ML) without quality data ML models cannot learn, perform or make accurate predictions.

- Data provides the examples from which models learn patterns and relationships.
- High-quality and diverse data improves how well models perform and generalize to new situations.
- It helps models to understand real-world scenarios and adapt to practical uses.
- Features extracted from data are important for effective training.
- Separate datasets for validation and testing measure how well the model works on unseen data.
- Data drives continuous improvements in models through feedback loops.

1.6 Advantages of Machine Learning

1. **Enhanced Efficiency and Automation:** ML automates repetitive tasks, freeing up human resources for more complex work. This leads to faster, smoother processes and higher productivity.
2. **Data-Driven Insights:** It can analyze large amounts of data to identify patterns and trends that might be missed by people and help businesses make better decisions.
3. **Improved Personalization:** It customizes user experiences by tailoring recommendations and ads based on individual preferences.
4. **Advanced Automation and Robotics:** It helps robots and machines to perform complex tasks with greater accuracy and adaptability. This is transforming industries like manufacturing and logistics.

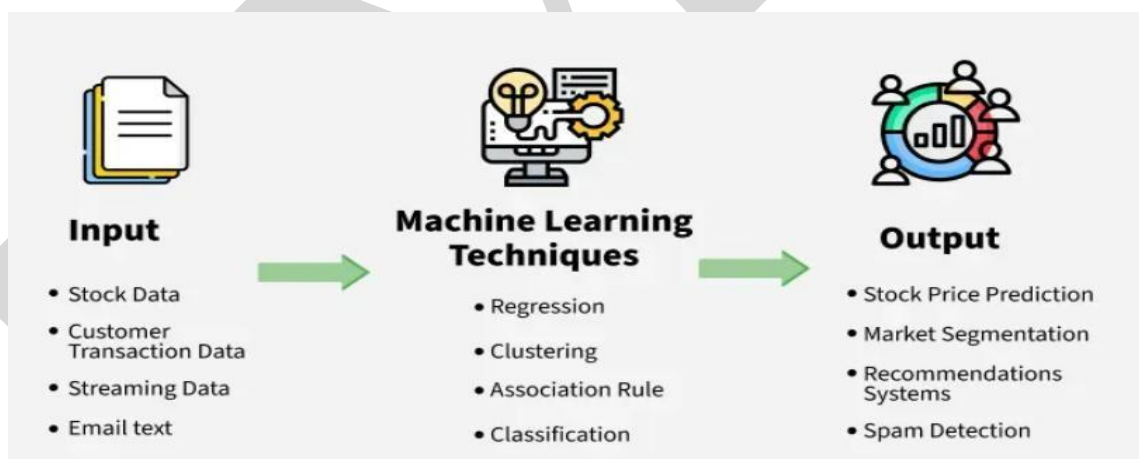
1.7 Challenges of Machine Learning

1. **Data Bias and Fairness:** ML models learn from training data and if the data is biased, model's decisions can be unfair so it's important to select and monitor data carefully.
2. **Security and Privacy Concerns:** Since it depends on large amounts of data, there is a risk of sensitive information being exposed so protecting privacy is important.
3. **Interpretability and Explainability:** Complex ML models can be difficult to understand which makes it difficult to explain why they make certain decisions. This can affect trust and accountability.
4. **Job Displacement and Automation:** Automation may replace some jobs so retraining and helping workers learn new skills is important to adapt to these changes.

1.8 Applications of Machine Learning

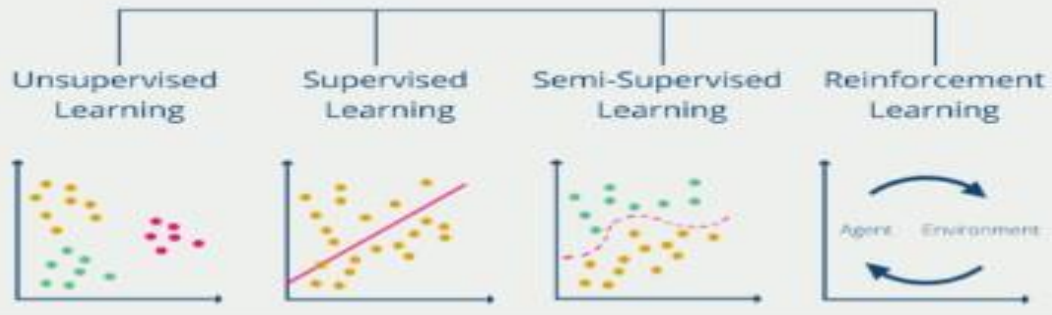
Machine Learning is used in many industries to solve problems and improve services. Here are some common real-world applications:

1. **Healthcare:** It helps doctors to diagnose diseases from medical images like X-rays and MRIs. It also predicts patient outcomes and personalizes treatments which improves healthcare quality.
2. **Finance:** In finance it detects fraudulent transactions in real time and supports algorithmic trading. It also helps to assess credit risk helps in making lending safer and faster.
3. **Retail and E-Commerce:** It helps in personalized product recommendations and forecasts demand to optimize inventory and also analyzes customer sentiment to improve shopping experiences.
4. **Transportation and Automotive:** Self-driving cars rely on ML to navigate and make decisions. It optimizes delivery routes and predicts vehicle maintenance needs which reduces downtime.
5. **Social Media and Entertainment:** Platforms like Netflix and YouTube use ML to recommend content we'll enjoy. It enables image and speech recognition for better user interaction.
6. **Manufacturing:** It improves quality control by detecting defects in products automatically and predicts machine failures in advance and helps in production processes.



2. Types of Machine Learning

Machine Learning



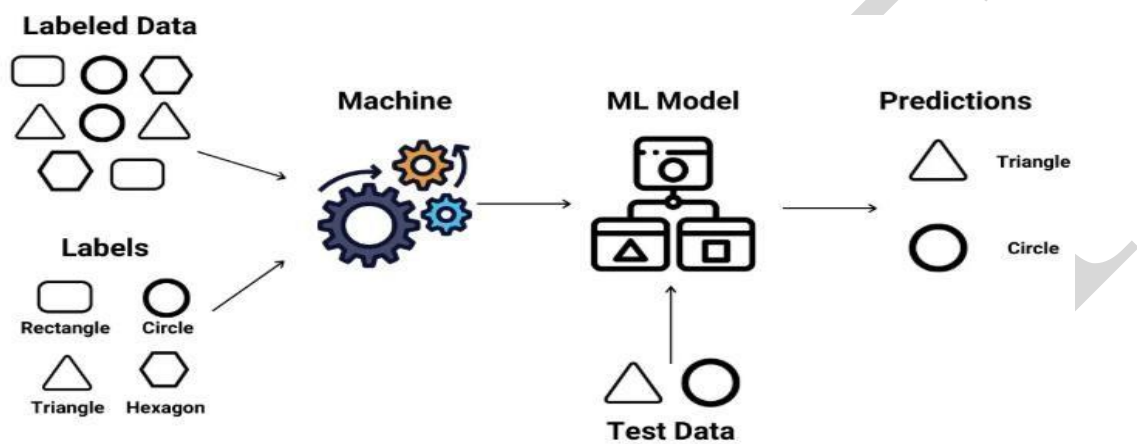
2.1 Supervised Learning

Learning from labeled data.

Tasks:

- Regression → Continuous output
- Classification → Discrete classes

Diagram:



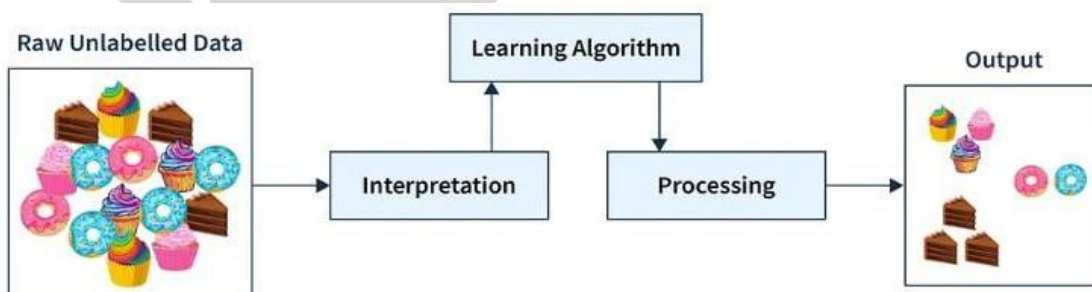
2.2 Unsupervised Learning

No labels; discovering hidden structure.

Tasks:

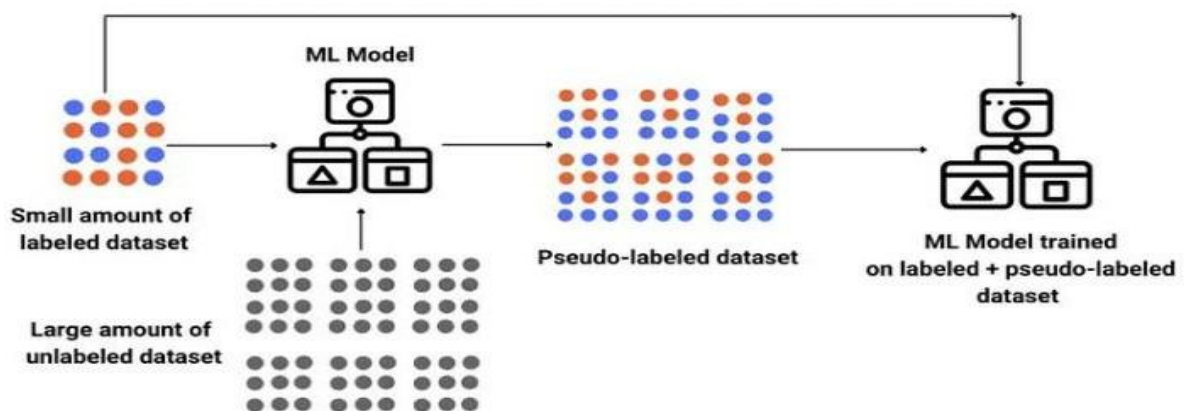
- Clustering
- Dimensionality reduction

Diagram:



2.3 Semi-Supervised Learning

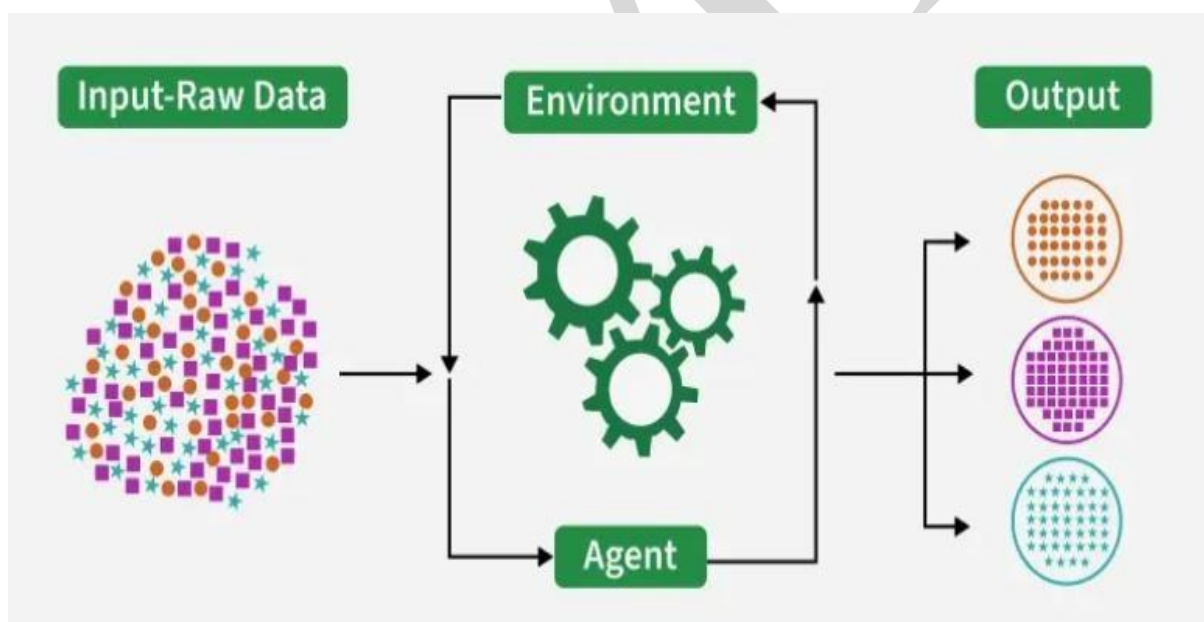
- Few labeled samples + many unlabeled samples



2.4 Reinforcement Learning

Learning via interaction with environment.

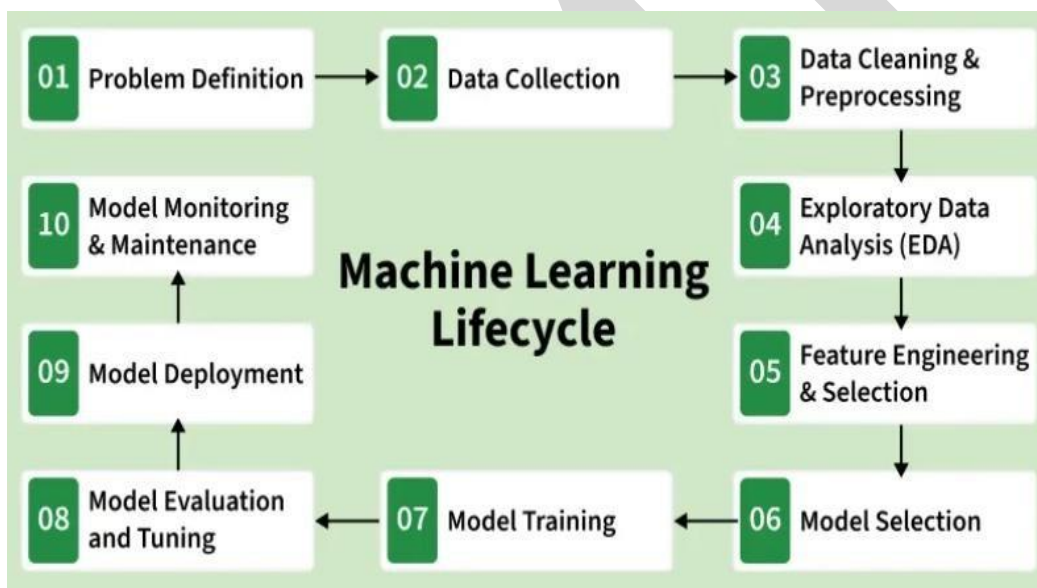
Diagram (Agent–Environment Loop):



Aspect	Supervised	Unsupervised	Semi-Supervised	Reinforcement
Data	Labeled	Unlabeled	Few labeled + many unlabeled	No labels
Goal	Predict output	Find patterns	Improve prediction with few labels	Learn best action
Feedback	Direct (labels)	None	Partial	Reward / Penalty
Output	Class / value	Clusters / features	Class / value	Policy (actions)
Examples	Spam detection	Customer grouping	Medical images	Games, robotics

3. Machine Learning Pipeline

3.1 End-to-End Workflow



Step 1: Problem Definition

Clearly define the business problem, objectives, scope, and success criteria by collaborating with stakeholders.

Step 2: Data Collection

Gather relevant, high-quality, sufficient, and diverse data required to solve the problem.

Step 3: Data Cleaning & Preprocessing

Handle missing values, outliers, inconsistencies, and transform data into a usable format.

Step 4: Exploratory Data Analysis (EDA)

Analyze data using statistics and visualizations to identify patterns, trends, and insights.

Step 5: Feature Engineering & Selection

Create meaningful features and select the most relevant ones to improve model performance.

Step 6: Model Selection

Choose appropriate algorithms based on problem type, data nature, and performance needs.

Step 7: Model Training

Train the model iteratively on historical data to learn patterns and minimize errors.

Step 8: Model Evaluation & Tuning

Evaluate using metrics (accuracy, precision, recall, etc.) and tune hyperparameters if needed.

Step 9: Model Deployment

Deploy the trained model into production systems for real-world use.

Step 10: Model Monitoring & Maintenance

Continuously monitor performance, detect drift, and retrain the model when required.

4. Machine Learning with Python

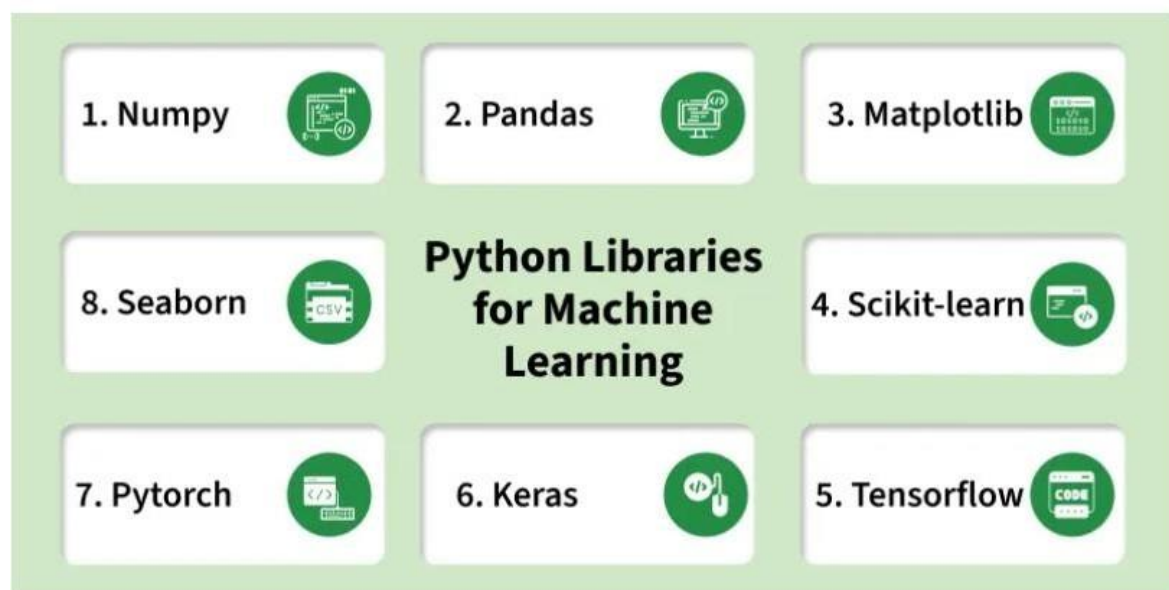
Python language is widely used in Machine Learning because it provides libraries like NumPy, Pandas, Scikit-learn, TensorFlow, and Keras. These libraries offer tools and functions essential for data manipulation, analysis, and building machine learning models. It is well-known for its readability and offers platform independence. These all things make it the perfect language of choice for Machine Learning.

Machine Learning involves building systems that can automatically learn patterns from data and make predictions or decisions without explicit programming. Python has emerged as the most widely used language for machine learning due to its simplicity, readability and its useful ecosystem of libraries. These libraries provide efficient tools for data handling,

visualization, feature engineering, model building and evaluation making the entire machine learning workflow faster and more reliable.

- They provide optimized implementations of complex algorithms
- They simplify data preprocessing and feature engineering
- They support rapid experimentation and prototyping
- They are widely used in both academia and industry

Some popular Python libraries for Machine Learning are:



1. NumPy

NumPy is a fundamental numerical computing library in Python that provides support for large, multi-dimensional arrays and matrices, along with a comprehensive collection of mathematical functions. In machine learning, NumPy is primarily used for handling numerical data, performing vectorized operations and implementing low-level mathematical computations efficiently.

- Used for numerical feature representation and transformation
- Enables fast mathematical operations through vectorization
- Serves as the computational backbone for many ML libraries
- Efficient memory management for large datasets

2. Pandas

Pandas is a high-level data analysis and manipulation library built on top of NumPy. It

introduces useful data structures such as DataFrame and Series, which allow machine learning practitioners to clean, transform and analyze structured data efficiently before feeding it into models.

- Used for data cleaning, transformation and preparation
- Handles missing, inconsistent and categorical data
- Simplifies exploratory data analysis
- Integrates seamlessly with ML and visualization libraries

3. Matplotlib

Matplotlib is a comprehensive data visualization library used to create static and interactive plots. In machine learning, it plays a critical role in understanding data distributions, detecting patterns and interpreting model performance through graphical representations.

- Used for visualizing datasets and model outputs
- Helps identify trends, skewness and imbalances
- Supports custom and publication-quality plots
- Essential for result interpretation

4. Scikit-learn

Scikit-learn is a widely used machine learning library that provides simple and efficient tools for classical machine learning tasks. It supports supervised and unsupervised learning algorithms along with preprocessing, model evaluation and validation utilities.

- Used for classification, regression and clustering
- Provides consistent and easy-to-use API
- Includes preprocessing and evaluation tools
- Ideal for traditional ML problems

5. TensorFlow

TensorFlow is a useful open-source deep learning framework developed by Google. It is designed for building, training and deploying large-scale neural networks and supports both research and production-level machine learning systems.

- Used for deep learning and neural networks

- Supports GPU and distributed training
- Highly scalable and production-ready
- Flexible model architecture design

6. Keras

Keras is a high-level neural network API that simplifies deep learning model development. It abstracts much of the complexity involved in building neural networks, making it especially suitable for beginners and rapid prototyping.

- Simplifies neural network creation
- Requires minimal code
- Supports both regression and classification
- Improves development speed

7. PyTorch

PyTorch is an open-source deep learning library known for its dynamic computation graph, which allows models to be modified during execution. This makes PyTorch highly flexible and popular in research and experimentation.

- Used for research-oriented deep learning
- Dynamic and intuitive model building
- Easier debugging and customization
- Supports custom training logic

8. Seaborn

Seaborn is a statistical data visualization library built on Matplotlib. It is designed to create informative and visually appealing plots that help in understanding relationships between variables during exploratory data analysis.

- Used for exploratory data analysis
- Works directly with pandas Data Frames
- Produces cleaner statistical plots
- Enhances data interpretation

5. Introduction to Data in Machine Learning



Data refers to the set of observations or measurements to train a machine learning models. The performance of such models is heavily influenced by both the quality and quantity of data available for training and testing. Machine learning algorithms cannot be trained without data. Cutting-edge development in Artificial Intelligence, automation, and data analysis is powered mostly by vast sets of data.

Types of Data in Machine Learning



Based on Structure

1. **Structured Data:** Tabular data, such as rows and columns, is used to organize and store structured data. Spreadsheets and databases frequently contain this type of data.

- **Examples:** Sales records, customer details, financial transactions.
- **Usage:** Useful in supervised learning tasks like regression and classification.

2. **Unstructured Data:** Processing unstructured data is more challenging because it lacks a preset structure.

- **Examples:** Text files, pictures, videos, and audio files are a few examples.
- **Usage:** Found in speech-to-text systems, image recognition, and natural language processing (NLP) applications.

3. Semi-Structured Data: This type of data falls somewhere between unstructured and structured data. It has organizational elements but does not fit nicely into a tabular format.

- **Examples:** JSON files, XML files, and NoSQL databases.
- **Usage:** Often used in web scraping, API responses, and social media analysis

Based on Representation

- **Numerical Data:** Features measured in numbers (e.g., age, income).
- **Categorical Data:** Represents Categories or labels (e.g., gender, fruit type).
- **Ordinal Data:** Categorical data with an essential order (e.g., clothing sizes: Small, Medium, Large).

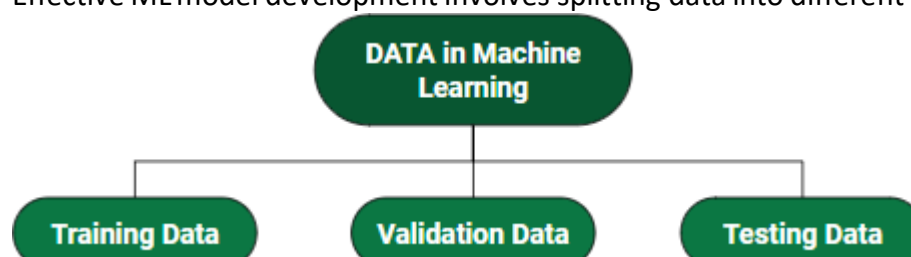
Based on Labeling

- **Labeled Data:** Includes input variables and corresponding target outputs. Example: Features like "age" and "income" with a label like "loan approval status."
- **Unlabeled Data:** Contains only input variables without any target labels. Example: Images without annotations.

Domain	Data Example
Healthcare	Patient records, lab results, imaging
Finance	Transaction logs, credit history
E-commerce	User reviews, purchase history
Transportation	GPS data, traffic reports
Social Media	Text, images, user engagement metrics

How do we split data in Machine Learning?

Effective ML model development involves splitting data into different sets:



1. Training Data

- Used to train the model.
- Model learns the patterns from this labeled data.

2. Validation Data

- Helps fine-tune the model by evaluating it during training.
- Useful for hyperparameter tuning and early stopping.

3. Testing Data

- Used after training is complete.
- Evaluates how well the model generalizes to unseen data

*In machine learning, **data is king**. Algorithms and models may be the engines, but data is the fuel. A deep understanding of data—not just its structure, but also how to prepare and use it effectively—sets the foundation for building powerful, reliable, and ethical machine learning systems.*

SUMMARY:

Note 1: Artificial Intelligence, Machine Learning and Deep Learning

Artificial Intelligence (AI) is a broad field of computer science focused on creating systems capable of performing tasks that normally require human intelligence such as reasoning, learning, perception, and decision-making.

Machine Learning (ML) is a subset of AI that enables systems to automatically learn patterns from data and improve performance without being explicitly programmed for every task.

Deep Learning (DL) is a further subset of ML that uses multi-layered neural networks to learn complex representations from large-scale data.

Relationship:

AI → ML → DL

- AI defines the goal (intelligent behavior)
- ML defines the approach (learning from data)
- DL defines the technique (deep neural networks)

Note 2: Traditional Programming vs Machine Learning

In traditional programming, rules and logic are explicitly written by humans. The system processes input data using fixed instructions to produce outputs.

In contrast, Machine Learning systems automatically learn rules from data. Instead of writing logic manually, data and expected outputs are provided, and the algorithm learns the mapping between them.

Key Difference:

Traditional Programming	Machine Learning
Rules are hard-coded	Rules are learned
Limited adaptability	Improves with data
Not scalable for complex tasks	Scales well with data
Human-driven logic	Data-driven logic

Machine Learning is especially effective when rules are complex, unknown, or constantly changing.

Note 3: Core Components of a Machine Learning System

Every Machine Learning system consists of three fundamental components:

1. Data

Data provides examples from which the system learns. The quality, quantity, and diversity of data directly impact model performance.

2. Algorithm

An algorithm is a mathematical method used to identify patterns in data. Different algorithms are designed for different tasks such as classification, regression, or clustering.

3. Model

A model is the learned representation obtained after training the algorithm on data. It is used to make predictions on new, unseen inputs.

The interaction of these three components enables a machine to learn and improve over time.

Note 4: Learning Process in Machine Learning

Machine Learning follows an iterative learning process:

1. Input data is provided to the system
2. The algorithm analyzes patterns in the data
3. A model is created and trained
4. Predictions are generated
5. Errors are measured using a loss function
6. Model parameters are updated to reduce errors
7. The process repeats until acceptable performance is achieved

This continuous improvement mechanism allows machines to learn from experience in a manner similar to human learning.

Note 5: Importance of Data in Machine Learning

Data is the foundation of Machine Learning. Without data, learning is not possible.

- High-quality data improves accuracy and reliability
- Diverse data helps models generalize better
- Labeled data enables supervised learning
- Unlabeled data supports unsupervised learning
- Validation and test data help evaluate real-world performance

Q&A

1. Why is Machine Learning preferred over traditional programming for problems like image recognition?

Answer:

Traditional programming requires explicitly written rules. In image recognition, defining rules for every possible variation of lighting, angle, background, and object shape is nearly impossible. Machine Learning overcomes this limitation by learning patterns directly from data.

For example, instead of manually programming what a “cat” looks like, an ML model learns from thousands of cat images and automatically identifies visual patterns. This makes ML scalable and effective for complex real-world problems where rules are not clearly defined.

2. A company wants a system that improves its performance automatically as more data

is collected. Which concept of Machine Learning enables this behavior? Explain.

Answer:

This behavior is enabled by the **learning process and feedback mechanism** in Machine Learning.

ML models are trained iteratively. As more data becomes available, the model updates its internal parameters to reduce errors and improve predictions. For example, Google Maps improves traffic prediction accuracy as more users share location data over time. This continuous improvement distinguishes ML from static traditional systems.

3. How does the availability of labeled data influence the choice of Machine Learning technique?

Answer:

The availability of labeled data directly determines the learning paradigm:

- **Labeled data available → Supervised Learning**
- **No labeled data → Unsupervised Learning**
- **Few labeled samples → Semi-Supervised Learning**

For example, email spam detection uses supervised learning because emails are labeled as spam or not spam. In contrast, customer segmentation in marketing often uses unsupervised learning because customer behavior data is not pre-labeled.

4. Consider a self-driving car learning to navigate traffic signals through rewards and penalties. Which type of Machine Learning is involved and why?

Answer:

This is an example of **Reinforcement Learning**.

In reinforcement learning, an agent interacts with an environment and learns by receiving rewards for correct actions and penalties for wrong ones. A self-driving car receives positive rewards for safe driving and penalties for traffic violations. Over time, it learns the optimal driving strategy through experience rather than labeled examples.

5. Why is data considered more important than algorithms in many Machine Learning applications?

Answer:

High-quality data enables models to learn accurate and meaningful patterns, even with simple algorithms. Poor data, on the other hand, can lead to inaccurate predictions

regardless of algorithm complexity.

For example, a simple logistic regression trained on clean, well-labeled medical data can outperform a complex deep learning model trained on noisy or biased data. Hence, data quality often has a greater impact on performance than algorithm selection.

6. A retail company wants to recommend products based on customer behavior without predefined categories. Which ML approach should be used and why?

Answer:

The company should use **Unsupervised Learning**, specifically **clustering**.

Unsupervised learning discovers hidden patterns without labeled outputs. By clustering customers based on browsing and purchase behavior, the system can identify groups with similar preferences. This approach is commonly used by e-commerce platforms like Amazon to generate personalized recommendations.

7. Explain how the Machine Learning pipeline helps in building reliable real-world systems.

Answer:

The ML pipeline provides a structured workflow that ensures reliability and consistency:

- Data collection ensures relevance and quality
- Data preprocessing improves usability
- EDA reveals patterns and anomalies
- Model training learns from historical data
- Evaluation ensures generalization
- Deployment enables real-world usage

For example, in healthcare diagnosis systems, following the ML pipeline ensures that predictions are accurate, validated, and safe before deployment in clinical environments.

8. Why is model evaluation on unseen data essential in Machine Learning?

Answer:

Evaluation on unseen data ensures that the model generalizes well and does not simply memorize training data (overfitting).

For instance, a student who memorizes answers without understanding will fail in a new exam. Similarly, an ML model must perform well on test data to be trusted in real-world applications like fraud detection or medical diagnosis.

9. A social media platform uses user interaction data to improve content recommendations daily. Which ML characteristics are demonstrated here?

Answer:

This system demonstrates:

- Data-driven learning
- Continuous improvement
- Personalized user experience

As users interact with content, the model learns preferences and updates recommendations accordingly. Platforms like YouTube and Instagram rely on this mechanism to keep content relevant and engaging.

10. How does the choice of programming language and libraries impact Machine Learning development?

Answer:

The choice of programming language and libraries affects development speed, scalability, and ease of experimentation.

Python is preferred because libraries like NumPy, Pandas, Scikit-learn, TensorFlow, and PyTorch provide optimized implementations of ML algorithms. For example, tasks that would take hundreds of lines in low-level languages can be achieved in a few lines using Python libraries, making experimentation and deployment faster and more reliable.

Case Study 1: Healthcare Diagnosis System

A hospital wants to develop a system that predicts whether a patient has a particular disease using past medical records and lab results. Each record is already labeled as “disease” or “no disease”.

Questions:

1. Identify the type of Machine Learning used.
2. Explain why this learning approach is suitable.
3. List two challenges the hospital might face while building this system.

Answer Points:

- **Type:** Supervised Learning

- **Reason:** Availability of labeled data
- **Challenges:** Data privacy, bias in medical data, model interpretability

Case Study 2: E-Commerce Recommendation Platform

An online shopping platform wants to recommend products to users based on browsing history and purchase behavior. There are no predefined categories or labels.

Questions:

1. Which Machine Learning technique should be used?
2. Name one algorithm category suitable for this task.
3. Explain how this improves user experience.

Answer Points:

- **Technique:** Unsupervised Learning
- **Algorithm Category:** Clustering
- **Improvement:** Personalized recommendations, increased engagement

Case Study 3: Self-Learning Traffic Signal System

A city plans to install an intelligent traffic signal system that adjusts signal timings based on traffic flow. The system receives rewards for reduced congestion and penalties for delays.

Questions:

1. Identify the learning paradigm involved.
2. Describe the role of rewards and penalties.
3. Mention one real-world benefit of this system.

Answer Points:

- **Learning Type:** Reinforcement Learning
- **Role:** Feedback mechanism to learn optimal actions
- **Benefit:** Reduced traffic congestion, improved fuel efficiency